

# Youngung Jeong

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## Education

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|--------------------------------------------------------------------------------|-------------------------|
| Hanyang University, BS, Department of Materials Science and Engineering        | 03/01/2001 – 02/29/2008 |
| POSTECH, MS, Graduate Institute of Ferrous Technology (supervisor: F. Barlat)  | 03/01/2008 – 02/28/2010 |
| POSTECH, PhD, Graduate Institute of Ferrous Technology (supervisor: F. Barlat) | 03/01/2010 – 02/28/2014 |

## Experiences

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|-------------------------------------------------------------------------|-------------------------|
| Assistant, associate, full professor, Changwon National University, ROK | 03/01/2017 – Present    |
| Faculty mentor guest, Los Alamos National Laboratory, NM, USA           | 10/07/2025 – 08/31/2026 |
| Short term visitor, Los Alamos National Laboratory, NM, USA             | 07/11/2024 – 08/16/2024 |
| Guest Scientist (offsite), Los Alamos National Laboratory, NM, USA      | 02/01/2022 – 02/1/2024  |
| Postdoc, POSTECH, ROK                                                   | 12/01/2016 – 02/28/2017 |
| Research Scientist, Clemson university, SC, USA                         | 03/01/2016 – 11/30/2016 |
| Postdoc, NIST, MD, USA                                                  | 03/01/2014 – 02/29/2016 |
| Research Affiliate, Los Alamos National Laboratory, NM, USA             | 04/01/2012 – 09/30/2012 |
| Guest Researcher, NIST, MD, USA                                         | 06/01/2011 – 12/31/2011 |

## Selected recent publications

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| • Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations<br><i>International Journal of Plasticity</i> 191<br>José Victoria-Hernández*, <u>Y. Jeong*</u> , Dietmar Letzig      | 2025 |
| • Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel<br><i>International Journal of Plasticity</i> 181<br>B. Jeon, S.-Y. Lee, J. Lee, <u>Y. Jeong*</u>                                           | 2024 |
| • A crystal plasticity finite element analysis on the effect of prestrain on springback<br><i>International Journal of Mechanical Sciences</i> , 237<br>M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, <u>Y. Jeong*</u> ,                                                | 2023 |
| • Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading<br><i>International Journal of Plasticity</i> 147<br><u>Y. Jeong*</u> , B. Jeon, C. N. Tomé | 2021 |

## Government-funded projects

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|------------------------------------------------------------------------------|-------------|
| Mid-career Researcher Program (PI), National Research Foundation of Korea    | 2023 - 2027 |
| BK21, National Research Foundation of Korea                                  | 2020 - 2027 |
| Engineering Research Center, National Research Foundation of Korea           | 2018 - 2024 |
| General Research Program (PI), National Research Foundation of Korea         | 2020 - 2023 |
| Virtual engineering platform project, Ministry of Trade, Industry and Energy | 2018 - 2020 |
| General Research Program (PI), National Research Foundation of Korea         | 2017 - 2020 |

## Skills

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Constitutive modeling: Phenomenological plasticity and crystal plasticity modeling

Languages: Korean, English

Computer skills: Linux, Git

Programming languages: Fortran, Python (NumPy, SciPy, and matplotlib), Matlab, shell scripts

Experimental Mechanics: uniaxial tension, shear, hydraulic bulge test, biaxial tests, digital image correlation (DIC)

Diffraction experiments: X-ray diffraction, Electron back-scattered diffraction (EBSD), Residual stress analysis

## GitHub repositories

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**Elasto-visco-plastic self-consistent model** (private repo) [github.com/youngung/evpsc](https://github.com/youngung/evpsc)

- Extended visco-plastic self-consistent model to account for elasticity ( $\Delta$ EVPC and  $\sigma$ EVPC)
- Stand-alone calculation capabilities
- Various example shell and python scripts as well as Jupyter notebook files
- User material subroutine (UMAT) of Abaqus/standard solver
- Written primarily in Fortran with Python and shell scripts

**VPSC8 and EVPC** (private repo)

[github.com/Jeonbohye/VPSC8-dEVPC](https://github.com/Jeonbohye/VPSC8-dEVPC)

- Combines VPSC8 and EVPC models
- Numerical implementation of VPSC,  $\Delta$ EVPC, and  $\sigma$ EVPC constitutive model
- Three separate build commands for executables of VPSC8,  $\Delta$ EVPC, and  $\sigma$ EVPC.

**In-house Python scripts for texture analysis**

[github.com/youngung/texture3](https://github.com/youngung/texture3)

- Can plot contoured pole figures and inverse pole figures from discrete orientations
- Written entirely in Python with open-sourced libraries including matplotlib, NumPy and SciPy
- Can generate RGB maps of EBSD (electron back-scattered diffraction) scans

## Misc.

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- Board member of Korean Society for Technology of Plasticity 2023-present
- Editorial board member of Korean Journal of Metals and Materials 2019-present

## Full list of publications

1. Application of  $\Delta$  EVPSC to interpret mechanical behavior of AZ31B sheet in warm temperature submitted  
Kyoungwon Hwang, Bohye Jeon, Jaeseong Lee, Hyuk-Jong Bong, Jinwoo Lee, Y. Jeong\*
2. Elasto-visco-plastic model for deformation of dual phase steels submitted  
Anirban Patra\*, Y. Jeong\*, Carlos N. Tome
3. A cluster-based mean-field crystal plasticity modeling framework to simulate dynamic recrystallization submitted  
Y. Jeong\*, Carlos N. Tome, Andrea Rovinelli, Laurent Capolungo
4. Calibrating self-consistent crystal plasticity models beyond stress-strain fitting: mechanism-activity features to discover solution families submitted  
Jinyeong Yu, Jaeseong Lee, Y. Jeong\*, T. Lee\*
5. Reverse engineering material behavior using Bayesian inference and finite element analysis on ring-pull test submitted  
B. Jeon, C. N. Tome, P. M. Beck, B. Eftinik, A. Talapatra, Y. Jeong\*, L. Capolungo\*
6. (1012) Twinning, Slip, and Texture in AZ31 Mg under Uniaxial Compression: A Comparative Study of Schmid Factor and Crystal Plasticity Modeling accepted  
*Metals and Materials International*  
Jongbin Go\*, Y. Jeong\*, Myeong-heom Park\*, Sangwon Lee, Si Gao, Nobuhiro Tsuji
7. Leveraging Machine Learning and Crystal Plasticity for Efficient Calibration of Yld2004-18p Anisotropic Yield Function 2026  
*International Journal of Mechanical Sciences*, 328, 112039  
A. S. Ebrahim, J. Kim, Y. Jeong, T. Park, H. Lim, B. L. Kinsey, J. Ha\*
8. Experimental and numerical investigations on the microstructure and tensile deformation behavior in flash-butt welded joints of boron steel and the effects of a post heating condition 2026  
*Materials Characterization*, 238, 116631  
Joonoh Moon\*, Gyuyeol Bae\*, Y. Jeong, Min-Ji Kwon, Jaeseong Lee
9. A Scientific Benchmark for Elasto-Plastic Constitutive Modeling - Part II: Blind Predictions, Calibration Strategies, and Benchmark Results 2026  
*International Journal of Material Forming*, 19:17  
Lorenz Maier, Erik Walz, Michael Hofmann, Joana Rebelo Kornmeier, Rongfei Juan, Lian Junhe, Youngeung Jeong, Bohye Jeon, Martin Milch, Alper Guner, Emad Maawad, Wolfram Volk, Christoph Hartmann
10. Role of recovery in the microstructure development and mechanical behavior of a ductile Mg-Zn-Nd-Y-Zr alloy: an analysis using EBSD data and crystal plasticity simulations 2025  
*International Journal of Plasticity*, 191, 104380  
José Victoria-Hernández\*, Y. Jeong\*, Dietmar Letzig
11. Modeling deformation, recovery, and recrystallization of tantalum using a higher order elasto-viscoplastic self-consistent model 2025  
*Journal of the Mechanics and Physics of Solids*, 194, 105925

- I. A. Riyad, B. Clausen, D. J. Savage, Y. Jeong, D. W. Brown, M. Knezevic\*
12. A critical discussion of elasto-visco-plastic self-consistent (EVPSC) models 2024  
*Journal of Materials Research Technology*, 33, 7596-7609  
 B. Jeon, Y. Jeong\*, C. N. Tomé
  13. Direct application of elasto-viscoplastic self-consistent crystal plasticity model to U-draw bending and springback of dual-phase high strength steel 2024  
*International Journal of Plasticity*, 181, 104098  
 Bohye Jeon, Shin-Yeong Lee, Jinwoo Lee, Youngung Jeong\*
  14. Thermal ratcheting of uranium simulated with a thermo-elasto-visco-plastic self-consistent polycrystal model 2024  
*Journal of Nuclear Materials*, 597, 155159  
Youngung Jeong\*, Carlos N. Tomé
  15. Crystal plasticity finite element simulations on extruded Mg-10Gd rod with texture gradient 2024  
*Journal of Magnesium and Alloys*, 12 , 3409-3430  
 Jaeseong Lee, Dirk Steglich, Youngung Jeong\*
  16. A comprehensive analysis of cermet design and thermal cyclic stability via elasto-viscoplastic crystal plasticity modeling 2024  
*International Journal of Plasticity*, 179, 104032  
 Glenn R Peterson, Youngung Jeong, Carlos N Tomé, Michael D Sangid\*
  17. Temperature-dependent behavior of CP-Ti interpreted via self-consistent crystal plasticity simulation 2024  
*Materials Science and Engineering: A*, 890, 145904  
 Bohye Jeon, Min-Su Lee, Tea-Sung Jun, Youngung Jeong\*
  18. Finite element analysis using elasto-visco-plastic self-consistent polycrystal model for E-form Mg sheet subjected to bending 2023  
*Journal of Magnesium and Alloys*, 11, 1393-1407  
 B. Jeon, M. S. Kim, S. H. Choi, Y. Jeong\*
  19. A crystal plasticity finite element analysis on the effect of prestrain on springback 2023  
*International Journal of Mechanical Sciences*, 237, 107796  
 M. Joo, M. -S. Wi, S.-Y. Yoon, S.-Y. Lee, F. Barlat, C. N. Tomé, B. Jeon, Y. Jeong\*
  20. Reconstructing orientation data from the images of IPF maps and ODF sections extracted from the literature: A data-collection method for machine learning 2023  
*International Journal of Plasticity*, 159, 103467  
 L. Kaushik, K.-S. Park, J.-G. Kim, J. Lee, Y. Jeong, S.-H. Choi\*
  21. Prediction and validation of stress triaxiality assisted by elasto-visco-plastic polycrystal model 2022  
*Korean Journal of Metals and Materials*, 60, 607-618  
 Jinhwa Park, Youngung Jeong\*

22. In-situ neutron diffraction study of lattice deformation behaviour of commercially pure titanium at cryogenic temperature 2022  
*Scientific Reports*, 12, 3719  
M.-S. Lee, T. Kawasaki, T. Yamashita, S. Harjo, Y.-T. Hyun, Y. Jeong, T.-S. Jun
23. Finite element analysis using an incremental elasto-visco-plastic self-consistent polycrystal model: FE simulations on Zr and low-carbon steel subjected to bending, stress-relaxation, and unloading 2021  
*International Journal of Plasticity*, 147, 103110  
Y. Jeong\*, B. Jeon, C. N. Tomé
24. An efficient elasto-visco-plastic self-consistent formulation: Application to steel subjected to loading path changes 2020  
*International Journal of Plasticity*, 135, 102812  
Y. Jeong\*, C. N. Tomé
25. Modelling-assisted description of anisotropic edge failure in magnesium sheet alloy under mixed-mode loading 2020  
*International Journal of Mechanical Sciences*, 181, 105680  
Y. Jeong\*, Dirk Steglich
26. Extension of the VPSC model to account for elasto-visco-plastic behavior using a perturbed viscoplastic approach 2019  
*Modelling and Simulation in Materials Science and Engineering*, 27, 085013  
Y. Jeong\*, C. N. Tomé
27. Superior tensile fracture strength of hot isostatically pressed TiC–steel metallic composite fabricated by a novel infiltration 2019  
*Materials Science and Engineering: A*, 764, 138260  
S. J. Park, Y. Jeong, C. W. Kim, J. H. Lee, S. C. Cho, S. B. Lee, S. K. Lee, D. H. Kim, H. U. Hong\*
28. Enhancement in viscoplastic self-consistent FLD prediction model and its application for austenitic and ferritic stainless steels 2019  
*Metals and Materials International*, 25, 1548–1563  
Y. Jeong\*, T. Manninen
29. A crystal plasticity model for describing the anisotropic hardening behavior of steel sheets during strain-path changes 2018  
*International Journal of Plasticity*,  
H. Kim, F. Barlat, Y. Lee, S. Zaman, C. S. Lee, Y. Jeong\*
30. A comparative study between micro- and macro-mechanical constitutive models developed for complex loading scenarios 2017  
*International Journal of Plasticity*, 93, 212-228  
Y. Jeong\*, F. Barlat, C. N. Tomé, W. Wen
31. Uncertainty in flow stress measurements using X-ray diffraction for sheet metals subjected to large plastic deformations 2016  
*Journal of Applied Crystallography*, 49, 1991-2004  
Y. Jeong\*, T. Gnäupel-Herold, M. Iadicola, A. Kreuziger

32. Texture-based forming limit prediction for Mg sheet alloys ZE10 and AZ31 2016  
*International Journal of Mechanical Sciences*, 117, 102-114  
D. Steglich, Y. Jeong\*
  
33. Forming limit prediction using a self-consistent crystal plasticity framework: a case study for BCC fiber textures 2016  
*Modelling and Simulation in Materials Science and Engineering*, 24, 055002  
Y. Jeong\*, M.-S. Pham, M. Iaidocola, A. Creuziger, T. Foecke
  
34. Multiaxial constitutive behavior of an interstitial-free steel: measurements through X-ray and digital image correlation 2016  
*Acta Materialia*, 112, 84-93  
Y. Jeong\*, M. Iaidocola, T. Gnäupel-Herold, A. Creuziger
  
35. Effect of martensitic phase transformation on the behavior of 304 austenitic stainless steel under tension 2016  
*Materials Science and Engineering: A*, 649, 174-183  
H. Wang\*, Y. Jeong\*, B. Calusen, Y. LiU, R. J. McCabe, F. Barlat, C. N. Tomé
  
36. Evaluation of biaxial flow stress based on Elasto-Viscoplastic Self-Consistent analysis of X-ray Diffraction Measurements 2015  
*International Journal of Plasticity*, 66, 103-118  
Y. Jeong, T. Gnaupel-Herold, F. Barlat, M. Iadicola, A. Creuziger, M.-G. Lee\*
  
37. Application of crystal plasticity to an austenitic stainless steel 2012  
*Modelling and Simulation in Materials Science and Engineering*, 20, 024009  
Y. Jeong\*, F. Barlat, M.-G. Lee
  
38. Biaxial Deformation Behavior of AZ31 Magnesium Alloy: Crystal-Plasticity-Based Prediction and Experimental Validation 2012  
*International Journal of Solids and Structure*, 49, 3551-3561  
D. Steglich\*, Y. Jeong, M. O. Andar, T. Kuwabara